



Food and Agriculture
Organization of the
United Nations



World Health
Organization

International Code of Conduct on Pesticide Management

Guidance note on addressing illegal trade of pesticides



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on Pesticide Management**

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illegal trade of pesticides**

Food and Agriculture Organization of the United Nations
World Health Organization
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Preface

The illegal trade in pesticides is a major global concern, particularly in low- and middle-income countries. This issue has implications on human health, environmental safety, agricultural integrity and economy. The Food and Agriculture Organization (FAO) and the World Health Organization (WHO) are actively working to solve the concern through a variety of projects and guideline publications.

This *Guidance note on addressing illegal trade of pesticides* was developed through the FAO/WHO Joint Meeting on Pesticide Management (JMPM). This guidance note is designed to strengthen legislative and regulatory frameworks that govern the protection of agriculture production, public health, the environment and global trade. It also provides regulatory and technical guidance to Member Nations and stakeholders, focusing on establishing frameworks for better oversight of pesticide distribution and retailing. By adhering to this guidance, Member Nations can protect their populations from risks associated with unregulated pesticide use. This guidance note also fosters international collaboration by providing a common set of standards and practices that countries can adopt. This cooperation is essential for monitoring trade flows, sharing information on illegal activities and enforcing regulations effectively.

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Abbreviations

FAO	Food and Agriculture Organization of the United Nations
GHS	Globally Harmonized System of Classification and Labelling of Chemicals
LMIC	low- and middle-income country
OECD	Organisation for Economic Co-operation and Development
POP	persistent organic pollutant
PPE	personal protective equipment
RAS	Rapid Alert System (OECD)
SDS	safety data sheets
JMPM	Joint Meeting on Pesticide Management
VOC	volatile organic compounds
WHO	World Health Organization

Declarations of interest

FAO and WHO reported that they had received and reviewed declarations of interest from all FAO and WHO expert panel members who participated in the 14th and the 15th JMPM and had concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meeting.

1. Understanding illegal trade of pesticides

Illegal trade of pesticides is an important concern in all regions of the world, posing serious threats to agricultural production, human health, the environment, and the economy. For the purposes of this guidance note, and in accordance with the *FAO/WHO International Code of Conduct on Pesticide Management*, the illegal trade of pesticides is defined as “any trade of pesticides (including biopesticides) that does not adhere to one or more legal obligations at the national level”.¹ From a practical perspective, this encompasses concerns related to: i) illegal pesticides; ii) illegal trade methods, including outlaw labelling and packaging; and/or iii) illegal markets. However, not all aspects of illegal trade of pesticides are appearing in all countries to an equal extent. As such, it is important to assess each country’s specific situation, as discussed further in [Section 3](#), taking into consideration a multisectoral One Health perspective (see [Box 1](#)).² Any actions at national level should be carefully targeted to address the country’s own domestic challenges, which may differ even among neighbouring countries. However, it should be noted that cooperation and coordinated action among countries provides the greatest and most efficient possibility to tackle illegal trade of pesticides.

1.1 Illegal pesticides

The Organisation for Economic Co-operation and Development (OECD) in its best practice guidance on this subject identifies an illegal pesticide as “Any pesticide which, for whatever reason, is not legal in the country of destination. This includes the subcategories of counterfeits, fakes, obsolete and unauthorised pesticides.”³ Other definitions of illegal or illicit pesticides exist in various literature and typically encompass obsolete or banned substances, unauthorized imports, and counterfeit labelling for fake, inactive or inappropriate ingredients as applicable in the country of use.^{4, 5}

Box 1. One Health

One Health is an integrated, unifying approach that aims to sustainably balance and optimize the health of people, animals and ecosystems. It recognizes that the health of humans, domestic and wild animals, plants, and the wider environment (including ecosystems) are closely linked and inter-dependent.

The approach mobilizes multiple sectors, disciplines and communities at varying levels of society to work together to foster well-being and tackle threats to health and ecosystems, while addressing the collective need for clean water, energy and air, safe and nutritious food, taking action on climate change, and contributing to sustainable development.

Source: FAO, WOA, WHO and UNEP. 2021. *Joint Tripartite (FAO, OIE, WHO) and UNEP Statement - Tripartite and UNEP support OHHLEP’s definition of “One Health”*.
<https://openknowledge.fao.org/server/api/core/bitstreams/54a0f96f-066e-4a16-a186-1a4bbd9d9303/content>

1.2 Illegal trade methods

Illegal traders commonly use the following techniques and practices:

- smuggling methods such as disguising products, splitting containers into small batches, separating labels from containers and chemicals, hiding/mixing pesticides with other commodities in the same container (e.g. consolidating goods), and using long logistic chains that multiply opportunities to change documents and for the people involved to mislead border controls. In countries where border controls do not exist, smuggling simply means transferring products across these porous borders;
- falsifying or adulterating shipment documents, for e.g. changing documents during transit, presenting fake certificates at transit entry points, and using safety data sheets (SDS) that do not correspond to the products, including:
 - exporting/importing under a different Globally Harmonized System of Classification and Labelling of Chemicals (GHS) code or group, for e.g. identifying the product as a veterinary drug or as a plant stimulant rather than a pesticide – therefore bypassing the pesticide registration route;
 - mislabelling (for e.g. no labels, wrong labels, fake labels, and labels with incomplete information);
- using wrong/unauthorized packaging and/or repackaging, such as:
 - re-using original empty pesticide containers, and refilling with counterfeit, unknown, or obsolete pesticides or decanting of pesticides into common beverage containers;
- placing seized illegal stocks (usually kept in temporary storage before being destroyed) back onto the market;
- increased use of online sales of illegally traded pesticides, which can consist of substandard and counterfeit pesticides, and/or products without the appropriate license or labelling, with delivery through courier companies;
- unauthorized blending (“garage” or “street pesticides”), with a decrease in the concentration of the active ingredients, or unauthorized mixing of registered products, or using other substitutes of adjuvants and inert ingredients which are not part of the registered formulation.

1.3 Illegal markets

Deliberate illegal trade of pesticides is often committed by organized criminal groups, as defined under international law in the *Convention Against Transnational Organized Crime and the Protocols Thereto*, 2004.⁶ The ongoing demand for pesticides, and an economic interest in illegal production and trade, combined with insufficiently deterrent penalties, are the main drivers of illegal pesticides appearing on the market. However, as highlighted in the UNEP/GRID-Arendal study on *Illegal Trade in Chemicals*, legal pesticides can be sold on the illegal market, and vice versa, illegally traded pesticides can be distributed through the legal market, and can take some of the following forms (p. 27):

- “Legal trade mixed with illicit goods, for example online marketing (legal) of sub-standard or counterfeit products.

- Illegal trade in illicit goods, such as smuggling to avoid taxes or to launder money, or intentionally placing legally produced chemicals on markets where they are restricted or banned.
- Illegal trade operations of illicit goods – illegally developed supply chains for illicit, mainly banned products.”⁷

1.4 Overall scale

The illegal trade of pesticides has been catalysed by the globalization of pesticide trade since the 1990s, and in recent years it has been enabled by the rapid growth of e-commerce and global crisis situations (e.g. the COVID-19 pandemic). By definition, the size of the illegal trade of pesticides is difficult to estimate. However, it is surmised that approx. one-third of pesticide trade is illegal in Africa^{8, 9, 10} and India,¹¹ approx. 24 percent in Brazil,¹² and between 10 percent and 14 percent in the European Union.^{5, 13, 14} Thus, the prevalence of illegal trade of pesticides can vary widely across countries.

Similarly, data on enforcement efforts are also hard to come by in many countries. However, some regions have recently seen an increasing number of law enforcement seizures of counterfeit, fake, sub-standard and unauthorized/unregistered pesticides, for example in the European Union during EUROPOL’s Silver Axe operations¹⁵ and in Brazil.¹² Furthermore, the European Union continues to detect residues of unknown chemicals in food imports,¹⁶ which are compromising food safety. In addition, counterfeit, sub-standard and fake vector control products have been found in the materials procured for public health vector control programmes, such as insecticide treated nets.¹⁷

1.5 Challenges for governments

Put together, the dimensions mentioned in [Section 1.4](#) make illegal trade of pesticides difficult to detect and stop. International shipments of illegally traded pesticides often escape oversight by regulators and customs due to the complexity and length of the supply chains; limited communication among national and international authorities; lack of knowledge in the modus operandi of illegal traders; the experience and know-how of criminal organisations; and the limited applicability of criminal or otherwise sufficiently deterrent sanctions.

The complexity of the supply chain is illustrated as follows:

- Manufacturing, formulation and use of a product (and components of the product such as active ingredients and adjuvants/solvents) often take place in different locations and countries.
- The ingredients of the final products can cross borders many times before reaching a final point of formulation and then the end-users.

Illegal traders take advantage of these long, complex, and difficult to trace systems, hence creating more opportunity for fraud at any weak points of entry into the regular supply chain. The result is a distortion of pesticide markets by replacing legitimate products with cheaper and possibly more hazardous or ineffective products. As such, illegal trade of pesticides can ultimately undermine the efforts (and credibility) of governments to reduce risks from pesticides.

2. Consequences of the illegal trade of pesticides

The illegal trade of pesticides can pose severe risks to human health, the environment, agricultural production and vector control, and for the economy. These risks include corresponding negative societal impacts through ineffective pest and disease management and unregulated chemical exposure. In particular, these can include:

2.1 Health impacts

- The unknown and potentially variable composition of illegally traded pesticides may have severe health impacts on users and other exposed persons, resulting in poisoning, death, or long-term/chronic effects. The first victims of illegal pesticides are often the users. In addition, because of behavioural patterns and limited awareness, children may face particular risks of poisoning and death.¹⁸
- Misdiagnosis: If poisoning (self-harm, intentional [i.e. suicide], or unintentional) occurs, any inaccurate, illegible or unavailable information about product contents can lead to misdiagnosis of the symptoms and their cause by healthcare providers. Lack of readily accessible healthcare and effective antidotes at rural health facilities, if any, may further increase risks to farmers, agricultural workers, vector control personnel and other persons exposed to the pesticides.
- Residues of unknown or unauthorized pesticides in food or water endanger food safety and may pose health risks for consumers. Should exposure occur, then self-care can be compromised due to the inaccuracy or lack of information about the composition of illegal pesticides. This will render useless any first aid instructions, and potentially make things worse where those instructions are inappropriate for the actual chemistry exposure.
- Occupational safety and health issues: unauthorized packaging (or illegal decanting of registered products into inadequate packaging) may fail to adequately provide relevant personal protective equipment (PPE) information and to protect users from exposure leading to adverse health effects.
- Increased incidence of vector-borne diseases: inadequate vector control due to use of substandard products (with inadequate efficacy) can lead to increased incidence of vector-borne diseases and failure of emergency vector control.
- Resistance among disease vectors: development of resistance may appear due to exposure to sub-optimal insecticide applications of substandard products and pesticide residues in the environment.

2.2 Environment impacts

- Biodiversity loss: The unknown and potentially variable composition of illegally traded pesticides may harm non-target species, including beneficial insects like bees, birds, aquatic life, and other wildlife. This can lead to a reduction in biodiversity and can disrupt ecosystems.¹⁹
- Soil degradation: Illegally traded pesticides may not be formulated correctly for agricultural use, and this can lead to soil contamination, reduce soil fertility, kill beneficial microorganisms, and disrupt soil ecosystems, leading to long-term soil degradation.²⁰

- Water pollution: Illegally traded pesticides may run off into nearby water bodies, with toxic effects on aquatic life, including fish kills and disruptions to aquatic ecosystems.²¹
- Air pollution: The use of illegally traded pesticides can lead to the release of volatile organic compounds (VOCs) and other harmful chemicals into the air, contributing to air pollution and potential adverse effects on respiratory health in humans and animals.²²
- Resistance development: The use of illegally traded pesticides can contribute to the development of resistance in pest populations. This can result in the use of increasingly potent pesticides to manage pests, perpetuating a cycle of escalating environmental harm and reducing the efficacy of existing pest control measures.²³
- Habitat destruction: The use of illegally traded pesticides can contribute to the death of plants and trees, further reducing habitats for wildlife and contributing to deforestation and desertification.²⁴

2.3 Economic and social impacts

- Farmers' livelihood can be harmed: farmers can suffer serious crop losses (through phytotoxicity), also livestock health impacts and contamination of the natural resources (soil, water, loss of pollinators) they rely on for their farming activities, in addition to the increase in the cost of crop insurance (when they can afford and access it).
- Low-quality or poorly adapted pesticides can result in inadequate pest control and low crop yields, and endanger food safety and security.
- Governments may lose a significant source of taxes, if untaxed illegally traded pesticides replace taxed legal products.
- Vector control programmes may spend more to rotate or replace with alternative and often costlier products to manage insecticide resistance.
- Increased disease incidence (or lack of reduction of disease incidence) cause increased health spending by people and health programmes.
- Illegally traded pesticides can harm fair markets and create issues related to unfair competition among sellers. Legitimate pesticide manufacturers can suffer financial loss because of dishonest competition and violation of intellectual property rights, as well as negative publicity with impacts on their reputation.
- Detection of residues of unknown illegally traded pesticides in food can result in rejection of food export consignments, as well as in the recall of food produced locally, with consequences for international trade. This may lead to a full destruction of the crops.

3. Assessing the current situation at the country level

As a starting point, beyond recognizing the general widespread challenges of porous borders and complex supply chains, countries need to assess their current national situation and identify their own priority challenges for attention and action. For example, the situation will be quite different in countries where regulations, inspections and border controls are weak, infrequent or non-existent, versus countries where pesticides are strictly regulated, inspections more frequent and border controls stringent. Depending on country context, the priority challenges related to the illegal trade of pesticides could potentially include:

- inadequate national pesticide legislation (including waste legislation) or inefficient implementation, compliance and enforcement of existing legislation;
- weak collaboration at the national level between authorities in charge of agriculture, health and environment, custom services, national and local inspection units, police departments and other regulatory and enforcement authorities;
- weak inspection, monitoring and control services, notably for post-registration inspection across the different ministries involved. Poor compliance and enforcement of the legislation by public and private actors, including prosecutors and the judiciary. Porous borders and weak import and export control services;
- insufficient procedures for data collection, reporting and sharing, and for integrated monitoring and control across different ministries and institutions;
- insufficient laboratory facilities for pesticide analysis and quality assessment including facilities and expertise specific for biopesticides;
- inadequate criminal or otherwise sufficiently deterrent sanctions for illegal activities and ineffective implementation of sanctions;
- addressing corruption and the accountability of public officials and organizations;
- unclear status of seized stocks and unclear responsibilities for the cost of destruction of seized stocks;
- ineffective engagement with the private sector in contributing to monitoring and controlling pesticide quality in the supply chain; and
- ineffective cooperation or coordination at the global and regional level.

Alongside understanding the country's priority challenges, it is important to understand and engage the main relevant stakeholder groups in the country. Depending on country context, the following stakeholders could potentially have a role throughout the pesticide life cycle and in addressing the illegal trade of pesticides:

Private sector

- Manufacturers, including stewardship, corporate social responsibility and audits, sales departments, etc.
- Manufacturers' associations
- Exporters and importers
- Distributors
- Sellers (wholesale and retail)

- Packaging/labelling companies
- Logistics companies, transporters/shipping companies
- Waste companies
- Private testing laboratories

Public sector

- Authorities (e.g. pesticide registration, enforcement, patent checks, finance, tax, customs, police)
- Governmental testing laboratories
- Extension services
- Health programmes, doctors, poison testing centres

Users

- Farmers (professional and non-professional)
- Pest control operators
- General public (purchasers of household products)
- Farmers' and workers' unions
- Government

4. Taking action to address the illegal trade of pesticides

After assessing the current country situation, it is necessary to identify options and take action. Lessons learned from across regions suggest that actions to counter the illegal trade of pesticides are more likely to be successful if:

- based on a foundational understanding that illegal trade of pesticides is a serious problem which may be linked to organized crime groups;
- carried out both internationally and nationally;
- competent government authorities work in close cooperation with legitimate industry;
- different available and efficient options are taken into consideration across all stages of the pesticide life cycle, as recommended by the *Code of Conduct*¹ and the *OECD Best Practice Guidance to Identify Illegal Trade of Pesticides*;²⁵ and
- conducted alongside a fully-fledged public awareness campaign and engaging with authorities, farmers, industry, and the public.

With these guiding principles in mind, this section presents several options for actions to address the illegal trade of pesticides. Depending on country context, these actions could include one or more of the following:

1. **Raise awareness of the illegal trade of pesticides among authorities**, farmers, industry, licensed operators (sellers, importers, commercial applicators), and the public to increase understanding that this is a serious problem that may be linked to criminal activities and that may have very serious health and environmental consequences, including on agriculture production.
 - Initiate dedicated campaigns and trainings for authorities, industry, farmers, and the public.
 - Stimulate responsible trade between buyers and sellers of pesticides along the entire supply chain, e.g. apply the know-your-customer principle.
 - Build the capacity of poison testing centres and ensure protocols to describe and, if possible, photograph the substance that caused the harm.
2. **Strengthen cooperation between all stakeholders** to encourage traceability, facilitate identification, withdrawal from the market, and safe disposal of illegal pesticides along the full pesticide life cycle.
 - Improve coordination among different national and subnational authorities and between these and other segments of society, including by adopting a One Health approach to pesticide and waste management.²⁶ Encourage the consultation and participation of end users, the industry, licensed pesticide operators and other stakeholders in decision-making.
 - Ensure a regular, coordinated screening of a pesticide's true identity, quality (i.e. organismal fitness) and efficacy at all steps along the value chain – covering both imported and domestically produced materials.
 - Improve import controls and restrictions to facilitate traceability and quality control.
 - Engage with inspectors, prosecutors, the judiciary and other enforcement authorities to improve their understanding of the risks associated with illegally traded pesticides.

3. **Improve awareness of, and (if possible) the practical availability of, alternatives to illegally traded pesticides**, including where appropriate, integrated pest management, and legally registered pesticides.
4. **Assess, and where needed, strengthen national pesticide legislation**, including waste legislation, in line with the *Guidance on Pesticide Legislation*.²⁷ In the context of illegal trade of pesticides, it is particularly important that legislation should:
 - Provide adequate inspection and enforcement powers across the whole pesticide life cycle.
 - Underpin a strong pesticide registration system alongside requirements for labels, containers, etc.
 - Impose points of regulatory control through registration or licensing of key operators like pesticide importers, manufacturers, formulators, sellers, etc.
 - Enable cooperation between relevant authorities, including for example, the competent authority for pesticide registration and customs authorities, for e.g. by adopting a Memorandum of Understanding and/or assigning dedicated contact points.
 - Assign a mandate to carry out post-registration market surveillance.
 - Enable information collection through record-keeping and reporting requirements imposed on key operators.
 - Clarify the status of seized stocks of illegally traded pesticides and which parties are responsible for costs associated with their storage and destruction.
 - Establish deterrent and effective applicable sanctions for legal violations, such as license revocation and monetary fines.
5. **Strengthen and modernize compliance and enforcement approaches and monitoring mechanisms**. In particular:
 - Establish the integration of traditional checks and digital technologies (see [Box 2](#)) to curb the illegal trade of pesticides.
 - Provide specialized training to inspectors, lawyers, prosecutors, judges and other enforcement authorities.
 - Increase effective, targeted surveillance at borders.
 - Train concerned personnel in custom authorities, pesticide enforcement departments, including all authorities with a role in the pesticide life cycle, alongside other enforcement authorities, such as prosecutors and the judiciary.
 - Emphasize the role of customs in applying international conventions for imports and exports (for example, the prior informed consent procedure).
 - Develop strategies for securing funding and resources. Depending on context, this could involve international funding, public–private partnerships, and allocation of national budgets.
 - Improve coordination among different ministries and stakeholders under a One Health approach.
 - Establish a robust data collection system to monitor the scale of the illegal trade of pesticides, identify trends, and evaluate the effectiveness of interventions. This could

include creating a centralized database for incidents of illegal trade and pesticide-related harm.

- Undertake continuous research and develop adaptation strategies to respond to new developments in the illegal trade of pesticides, such as the emergence of new chemicals or changes in trade routes.
- Strengthen ties with public health initiatives by collaborating with health agencies to monitor health impacts, provide healthcare training on pesticide poisoning, and integrate pesticide management into public health policies.

6. Strengthen global, regional, and national cooperation and collective efforts related to illegal trade of pesticides. Depending on country context, this could include to:

- Encourage the ratification and implementation of existing relevant international treaties, such as the *Rotterdam Convention on the Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade (Rotterdam Convention)*.²⁸
- Consider adopting the *OECD Recommendation of the Council on Countering the Illegal Trade of Pesticides*, which is open to all governments.²⁹
- Where applicable, promote regional cooperation and harmonization on pesticide registration and trade, including fast-track or proportional arrangements for low-risk substances and compounds.
- Develop and endorse common technical guidelines, aligned with those of OECD.
- Use existing networks and participate in the OECD Rapid Alert System (RAS) for suspected illegal international trade of pesticides, which is open to all governments.
- Apply the *WHO Guidelines for Procuring Public Health Pesticides* and encourage “responsible government procurement”,³⁰ including accountability of public officials and organizations.
- Depending on local context, consider the establishment of cross-border task forces and joint operations to tackle the illegal trade of pesticides in border areas.

Box 2. Digital technologies to curb illegal trade of pesticides

Digital technologies such as blockchain have the potential to provide end-to-end product traceability, chain of custody, and access to real time information about the products being traded. This could also increase the capacity to track and trace pesticide products, at the border and on online platforms. Such technologies, including big data analytics and artificial intelligence, also offer opportunities for better risk management at the border, like automated inspections of containers and cargos. However, traditional ways of monitoring are necessary to complement these technologies, especially in regions where such technologies have limited penetration. As mentioned in the Organisation for Economic Co-operation and Development (OECD) publication on digital technologies (Frezal and Garsous, 2020), “their adoption is nevertheless challenging given the large upfront costs associated with IT development and infrastructures, and their complexity. This is especially the case for developing countries.”³¹

Source: Frezal, C. & Garsous, G. 2020. *New Digital Technologies to Tackle Trade in Illegal Pesticides*. OECD Trade and Environment Working Papers 2020/02. <https://doi.org/10.1787/9383b310-en>

FAO and WHO welcome readers' feedback

FAO and WHO welcome readers' feedback on use of these guidance notes. FAO and WHO consider that these guidance notes are a living document that could be further improved. They therefore particularly value any feedback from users of the guidance and welcome any comment. They also value examples of how the guidance notes were used.

Please send your suggestions, comments and examples to pesticide-management@fao.org indicating the title of the guidance and the relevant section and page.

Notes

- ¹ FAO (Food and Agriculture Organization of the United Nations) & WHO (World Health Organization). 2014. *The International Code of Conduct on Pesticide Management*. Rome. <http://www.fao.org/3/I3604E/i3604e.pdf>
- ² FAO, WOA, WHO and UNEP. 2021. *Joint Tripartite (FAO, OIE, WHO) and UNEP Statement - Tripartite and UNEP support OHHLEP's definition of "One Health"*. <https://openknowledge.fao.org/server/api/core/bitstreams/54a0f96f-066e-4a16-a186-1a4bbd9d9303/content>
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The International Code of Conduct on Pesticide Management provides countries with a framework for supervision over pesticides within their territories. The illegal trade of pesticides poses significant risks to public health, environmental safety, and agricultural integrity. The Food and Agriculture Organization of the United Nations (FAO) has established the Code of Conduct on the Distribution and Use of Pesticides to address these issues through the development of the *Guidance note on addressing illegal trade of pesticides*. This guidance note promotes safe practices in pesticide distribution and use, ensuring only approved products are available on the market. It provides concise technical guidelines to elaborate specific articles of the Code of Conduct. It is specifically designed for the regulatory authorities, distributors, retailers and end users involved in trading of pesticides. Illegal pesticides undermine legitimate businesses and disrupt fair competition within the agrochemical industry. This guidance note helps establish a level playing field by promoting compliance with regulations and encouraging distributors and retailers to implement comprehensive strategies that not only address illegal activities but also promote safer agricultural activities.

It is a global issue linked to organized crime groups and requires coordinated efforts across borders. This guidance note fosters international collaboration by providing a common set of standards and practices for countries to adopt, essential for monitoring trade flows, sharing information about illegal activities, and enforcing regulations effectively. This guidance note also provides a One Health strategy which is crucial in addressing the various difficulties posed by the illegal pesticide trade. The guidance note also provides mitigation strategies to tackle this issue, with various options that should be considered, including raising awareness, initiating campaigns, promoting responsible trade, building poison testing centres, strengthening cooperation, improving coordination, screening pesticides, modernizing compliance, securing funding, establishing a robust data collection system, conducting research, strengthening ties with public health initiatives, and strengthening global, regional, and national cooperation.

